

SANDEEPANI SENEVIRATHNA

Senior Software Engineer — C#/.NET Windows Desktop, POS & Device Integration

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Professional Summary

Senior Software Engineer with 15+ years of experience architecting and maintaining enterprise software using **C#, .NET, WPF, and SQL Server**. Strong background in building **Windows desktop applications, POS solutions, and hardware-integrated workflows** (POS terminal integration, Wi-Fi-direct printing, thermal printing, webcam and barcode readers). Experienced in debugging complex application-level issues involving memory leaks, thread locks, and handles. Adept at bridging managed .NET code with external systems via P/Invoke. Comfortable working across the full SDLC in Agile/Scrum environments, utilizing Git and release management tooling. Also familiar with **Visual Studio Diagnostic Tools** and Docker basics.

Core Technical Skills

- **Desktop & Architecture:** C#, .NET Framework/Core 3/.NET 5+, WPF, MVVM, Windows Services, P/Invoke, Multithreading.
- **Hardware & Integration:** POS Systems and terminal integration, webcam integration, Barcode reading, Wi-Fi- Direct Printing, Kiosk APIs, Device Synchronization.
- **Debugging & Diagnostics:** Visual Studio Diagnostic Tools, Memory Leak Investigation, Thread Lock Analysis, Handle/Resource Profiling.
- **Backend & Data:** ASP.NET Web API, WCF, Entity Framework, SQL Server, SQLite, MongoDB Realm, LINQ.
- **Tools & Delivery:** Git, BitBucket, JIRA, Agile/Scrum, Docker (Basics), development and Mobile App Store Releases, Release Documentation, Code Reviews.
- Currently upskilling in C++/Win32 and SDK-level component development to transition deeper into Windows device ecosystems.

Professional Experience

Senior Software Engineer | Ecological Ltd. (2019 – Present)

Ecological builds POS, ERP, and hardware-integrated solutions for UK-based retail, wholesale, hospitality and religious organization clients.

- **Windows Desktop & POS Architecture (TempleTrack, Neutri Neuralage):** Maintained C# WPF desktop applications for POS billing, inventory, and back-office management. Utilized MVVM principles and Windows RDLC reports to deliver stable, high-performance code and better UI experiences for end-users.
- **Hardware & Device Integration (HHD, Witmeg REPS, TempleTrack):** Integrated apps with external hardware including Wi-Fi-Direct protocol for field sales POS invoice printing (Witmeg REPS) and POS/webcam integrations for TempleTrack (Windows). Also maintained an Android based barcode-reader/inventory management application (HHD).
- **Backend, Service Bus & Kiosk APIs (TempleTrax, EMessages, CloudSync):** Created ASP.NET Web API and Entity Framework backend services for self-service kiosks (TempleTrax), handling ticket transactions, print templates, and cloud synchronization via Node.js/Windows Services.
- **Debugging & Performance Optimization:** Investigated and resolved complex defects involving data synchronization, UI thread locks, and memory usage, ensuring crash-free stability for mission-critical POS terminals.
- **Cross-Platform & Real-time Sync:** Upgraded mobile field apps to utilize MongoDB Realm for real-time offline-first data synchronization with cloud databases and REST API services.
- **Agile Delivery & Mentorship:** Participated in code reviews, guided junior developers, and conducted releases and supported testing teams within JIRA/BitBucket Agile workflows and supported application support teams.

Software Engineer | Bileeta (Pvt) Ltd. (2014 – 2019)

Bileeta provided multi-tenant SaaS ERP and supply chain solutions for agriculture, manufacturing, and healthcare.

* **Enterprise Architecture (Entution ERP):** Developed transaction, master, and configuration modules for a 3-tier service-based ERP system using C#, ASP.NET Web API, and SQL Server with custom jQuery plugins on the frontend. * **Database & Reporting:** Optimized complex stored procedures and database scripts in MS SQL Server. Developed and maintained financial and operational reports using Crystal Reports and RDLC. * **Supply Chain Logistics (Entution Vesta):** Contributed to a cloud-based pharmaceutical logistics system for the Nepali Health Ministry (via USAID), focusing on procurement, inventory, and warehousing modules. * **Performance Enhancements:** Conducted UI bug fixing, JavaScript/jQuery optimizations, and backend troubleshooting to improve system responsiveness across cloud and on-premise deployments.

Associate Software Engineer | Duo Software (2011 – 2014)

- **SaaS Billing & CRM (BillUsers):** Developed modules for subscription management, CRM, and tax calculations for a cloud-based Pay-TV subscriber billing system using ASP.NET MVC, C#, and SQL Server.
- **POS Terminal Form Development:** Built the Sales Terminal module for a derived Point of Sales system, contributing to database design, and black-box/unit testing.

Earlier Career (2010 – 2011)

- **Software Engineer, Centura Solutions:** Developed PHP/MySQL taxi rental management systems and resolved web security/SEO issues for UK clients.
- **Trainee Web Developer, FlexSoft:** Assisted in the development of a 3-tier PHP/MySQL Vehicle Fleet Management System and banking system.

Education

- **MSc in Information Technology** | University of Moratuwa (Completed 2017)
- **BSc (Honours) in Computing (First Class)** | University of Portsmouth, UK / SIC (Completed 2010)
 - *Award: Best Collaborative Student Project (Archaeological Data Management System using C#, WCF, and SQL Server).*

Relevant Certifications & Training

- **Certificate on Programming in C#** — Microsoft / NetAssist International (2018)
- **Competent Scrum Master Certification** — Project Management Solutions (2019)
- **Certified Kubernetes Administrator (CKA) with Docker** course — WinIT Solutions (2022)